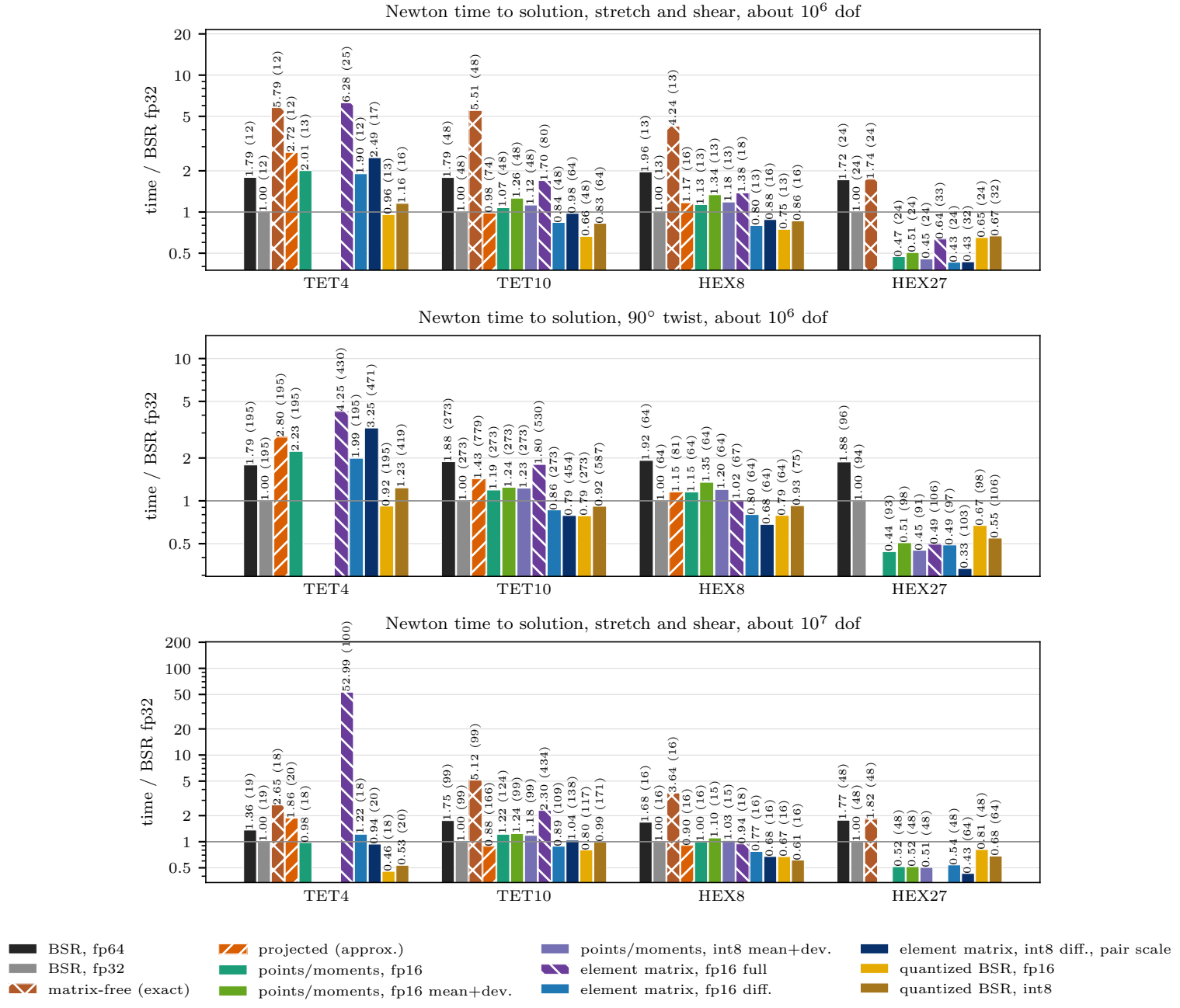

Newton time to solution, stretch and shear, about 10^6 dof
time / BSR fp32
20
10
5
2
1
0.5
1.79 (12)
1.00 (12)
5.79 (12)
2.72 (12)
2.01 (13)
6.28 (25)
1.90 (12)
2.49 (17)
0.96 (13)
1.16 (16)
1.79 (48)
1.00 (48)
5.51 (48)
0.98 (74)
1.07 (48)
1.26 (48)
1.12 (48)
1.70 (80)
0.84 (48)
0.98 (64)
0.66 (48)
0.83 (64)
1.96 (13)
1.00 (13)
4.24 (13)
1.17 (16)
1.13 (13)
1.34 (13)
1.18 (13)
1.38 (18)
0.80 (13)
0.88 (16)
0.75 (13)
0.86 (16)
1.72 (24)
1.00 (24)
1.74 (24)
0.47 (24)
0.51 (24)
0.45 (24)
0.64 (33)
0.43 (24)
0.43 (32)
0.65 (24)
0.67 (32)
TET4
TET10
HEX8
HEX27
Newton time to solution, 90° twist, about 10^6 dof
time / BSR fp32
10
5
2
1
0.5
1.79 (195)
1.00 (195)
2.80 (195)
2.23 (195)
4.25 (430)
1.99 (195)
3.25 (471)
0.92 (195)
1.23 (419)
1.88 (273)
1.00 (273)
1.43 (779)
1.19 (273)
1.24 (273)
1.23 (273)
1.80 (530)
0.86 (273)
0.79 (454)
0.79 (273)
0.92 (587)
1.92 (64)
1.00 (64)
1.15 (81)
1.15 (64)
1.35 (64)
1.20 (64)
1.02 (67)
0.80 (64)
0.68 (64)
0.79 (64)
0.93 (75)
1.88 (96)
1.00 (94)
0.44 (93)
0.51 (98)
0.45 (91)
0.49 (106)
0.49 (97)
0.33 (103)
0.67 (98)
0.55 (106)
TET4
TET10
HEX8
HEX27
Newton time to solution, stretch and shear, about 10^7 dof
time / BSR fp32
200
100
50
20
10
5
2
1
0.5
1.36 (19)
1.00 (19)
2.65 (18)
1.86 (20)
0.98 (18)
52.99 (100)
1.22 (18)
0.94 (20)
0.46 (18)
0.53 (20)
1.75 (99)
1.00 (99)
5.12 (99)
0.88 (166)
1.22 (124)
1.24 (99)
1.18 (99)
2.30 (434)
0.89 (109)
1.04 (138)
0.80 (117)
0.99 (171)
1.68 (16)
1.00 (16)
3.64 (16)
0.90 (16)
1.00 (16)
1.10 (15)
1.03 (15)
0.94 (18)
0.77 (16)
0.68 (16)
0.67 (16)
0.61 (16)
1.77 (48)
1.00 (48)
1.82 (48)
0.52 (48)
0.52 (48)
0.51 (48)
0.54 (48)
0.43 (64)
0.81 (48)
0.68 (64)
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
matrix-free (exact)
projected (approx.)
points/moments, fp16
points/moments, fp16 mean+dev.
points/moments, int8 mean+dev.
element matrix, fp16 full
element matrix, fp16 diff.
element matrix, int8 diff., pair scale
quantized BSR, fp16
quantized BSR, int8